

- 10 (a) (i) State what is meant by the luminosity of a star.

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..... [2]

- (ii) Explain how a standard candle in a distant galaxy can be used to determine the distance of the galaxy from an observer.

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..... [3]

- (b) The Sun has a radius of 6.96×10^8 m and a surface temperature of 5780 K.
Light from the Sun is observed to have a peak intensity at a wavelength of 501 nm.

- (i) Calculate the luminosity of the Sun. Give a unit with your answer.

luminosity = unit [2]

- (ii) Another star emits radiation that has a peak intensity at a wavelength of 624 nm.

Determine the surface temperature of this star.

surface temperature =K [2]

[Total: 9]